Coffee Chain Assignment
Tableau Visualization Case Study

Submitted to:

Professor Bin Li

Wright State University

Submitted by:

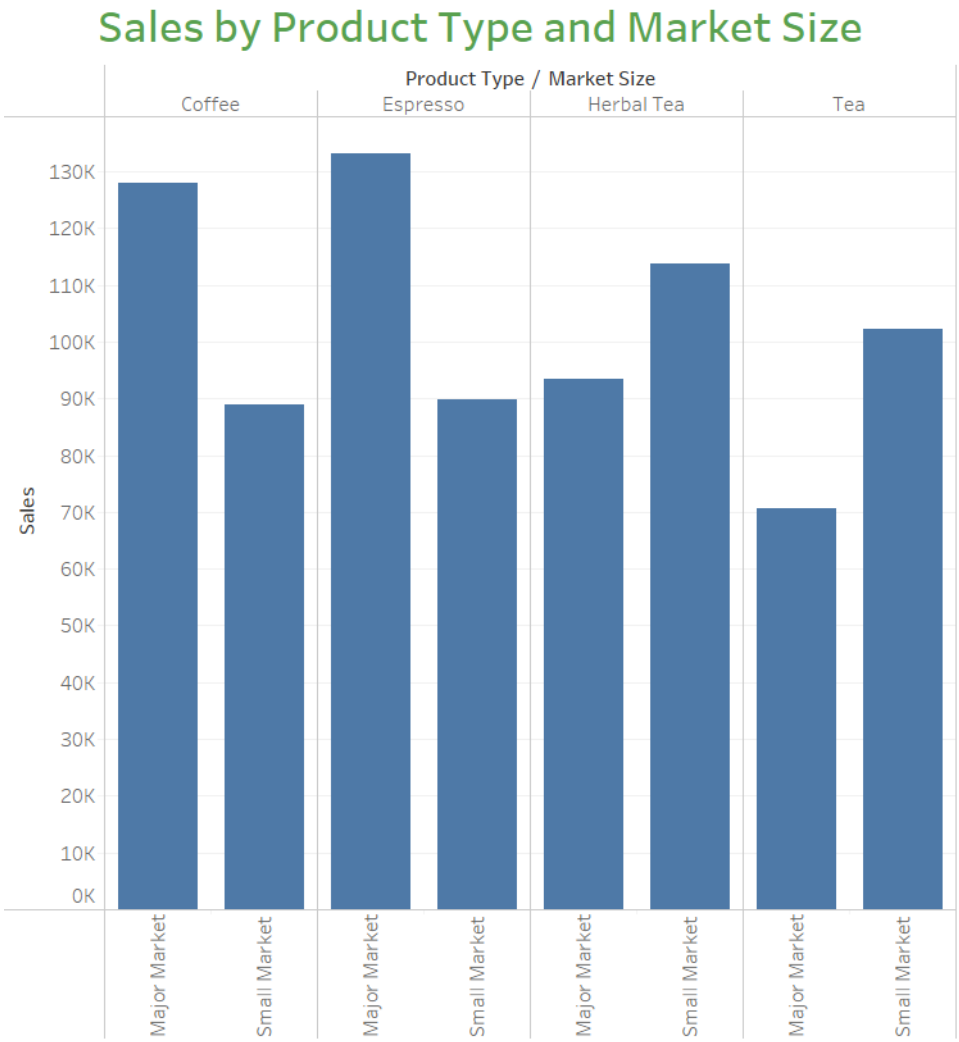
Sushant Bhatta

UID: U01105863

Coffee Chain Exercise

1) Your VP is interested in whether total sales by product type is impacted by market size? What is your conclusion? What type of chart did you use? Why?

It can be seen from the "Sales by Product Type and Market Size" bar chart that market size has an effect on overall sales for all product types. Coffee and espresso are two such products that have much more sales in large markets, with coffee selling almost 125K and espresso around 130K. The sales of coffee and espresso drop in small markets, however, with each falling below 95K. Herbal tea and tea, however, demonstrate a reverse trend: they do poorly in big markets, whose sales are still below 115K and 70K for them, respectively, while in small markets their sales are higher, with herbal tea exceeding 110K and tea exceeding 100K.



Sum of Sales for each Market Size broken down by Product Type.

A cluster bar chart, which in a good way compares sales statistics for each of the product categories across both sizes of markets side by side, was used. It is quite possible to see trends in the sales performance and comprehend how varying items react in relation to the market size as a result of this graphical orientation that makes the comparisons easy. It points out the necessity of distinct strategies for offerings in big vs small markets and proposes that market size is a determining factor for what products succeed.

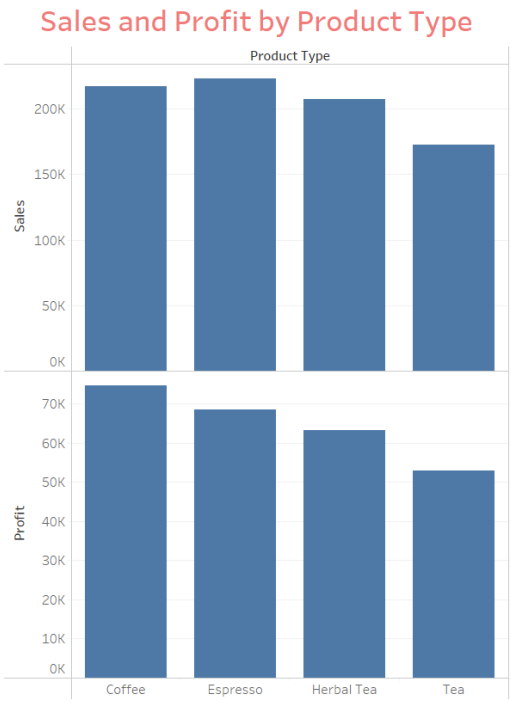
2) Which product types are doing better in small markets vs. large markets? Why do you think this is the case?

We can clearly see that small and big markets have different product performance as per the figures presented in the "Sales by Product Type and Market Size" graph. With sales of around 130K, large markets are led by espresso, followed by coffee with sales of around 125K. But in extremely small markets, where the two products sell fewer than 95K each, their sales take a serious dip. However, tea and herbal tea sales are more evenly distributed among market sizes. Herbal tea does especially well in small markets, with over 110K in sales versus less than 75K in large markets. Similarly, Tea sells approximately $100,000 in smaller markets, a high figure compared to its sales in large markets, which are less than $70,000.

The above trend indicates that tea and herbal tea work better in small markets, maybe because the city consumers there are health-conscious and price-sensitive and prefer traditional or healthy beverages. Coffee and espresso work better in large markets, maybe because the city consumers prefer specialty or premium beverages more. These findings underscore the importance of adapting product plans to different sizes of markets based on customers' behavior and desires.

3) How are total sales related to total profit by product type? Which product type has the highest profit? Highest sales?

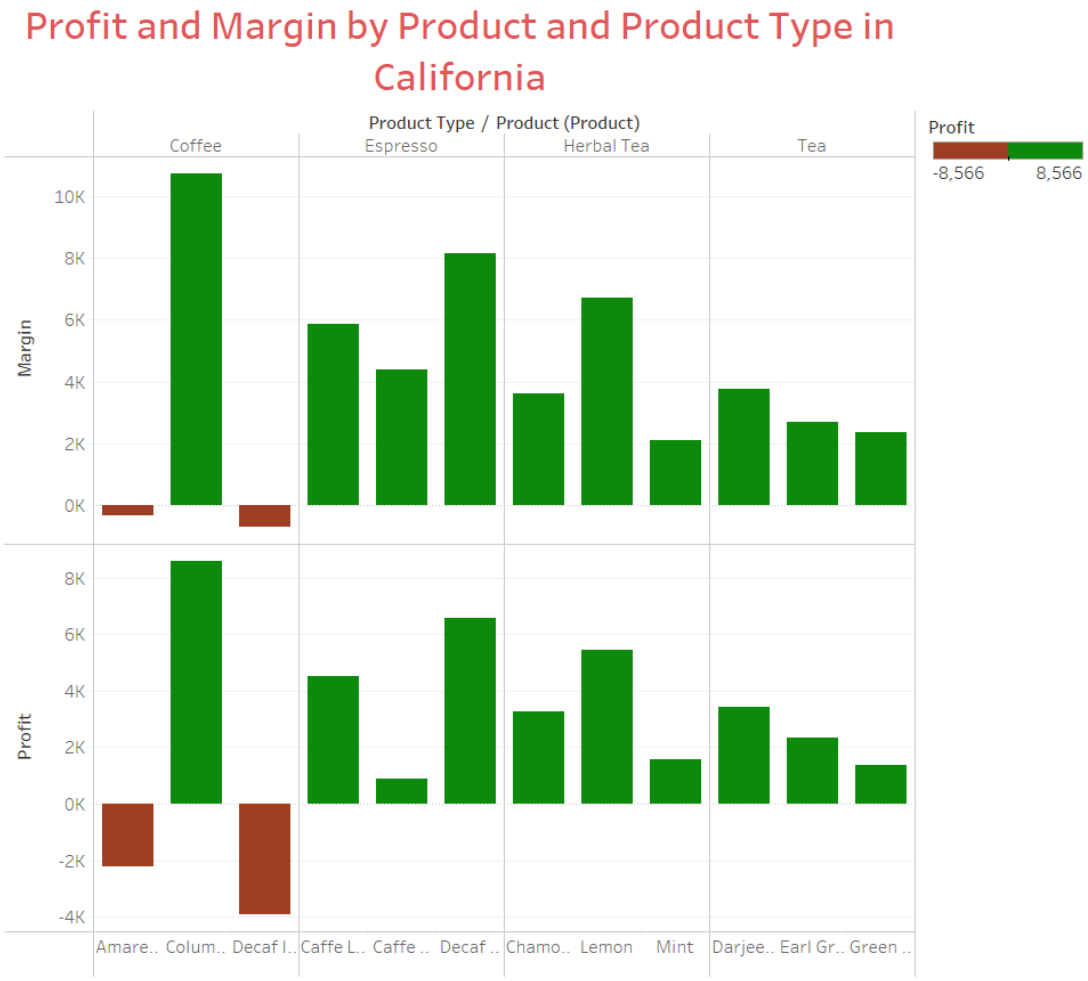
The "Sales and Profit by Product Type" graph indicates total profit and total sales have a positive relationship, with popular product types tending to make higher profits. The relationship is not strictly linear, though, indicating varying products have varying margins.



Coffee falls just behind Espresso in sales, but it leads on overall revenue at just over $70K. This would mean that out of all product groups, coffee would have the largest profit margin. Espresso has the largest overall sales (just over $220,000), but second in profit, so perhaps a less profitable per unit sold than coffee. Tea has the smallest numbers for both sales and profit, with herbal tea second.

4) You want to find out more about business in California. Show profit margin and profit by Product and Product type. In which product/type is the company losing money in California?

A more detailed look at the "Profit and Margin by Product and Product Type" graph in California shows that certain areas of the coffee category have clearly underperformed. While there are a few bright spots in the coffee product category as a whole, including Columbian Regular, Amaretto and Colombian Decaf have suffered significant losses. Besides losing over $3,000 to Colombian Decaf, both these items also have negative margins, i.e., either their selling price is below cost or they have very poor cost control. The red columns in the profit and margin section of the chart well corroborate the same.



Sum of Margin and sum of Profit for each Product (Product) broken down by Product Type. Color shows sum of Profit. The data is filtered on State, which keeps California.

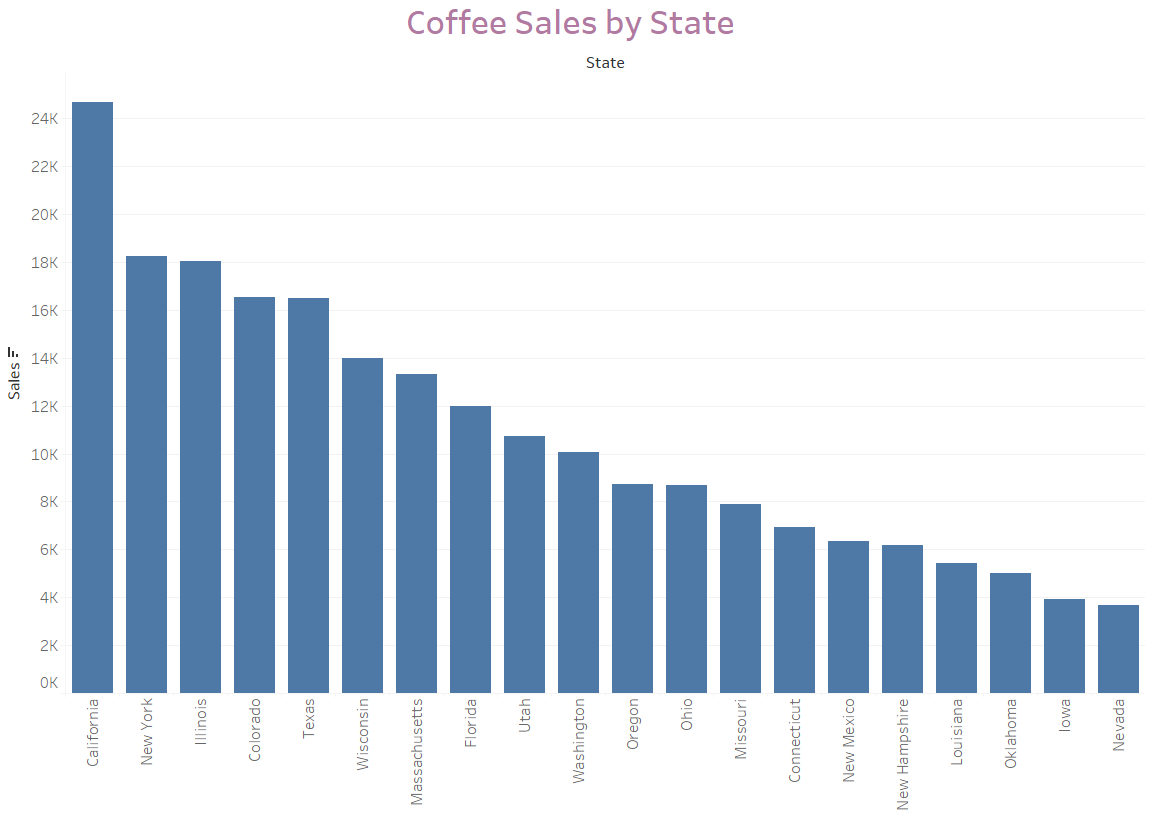
However, all of the products in the Espresso, Herbal Tea, and Tea categories are showing favorable margins and profitable results, indicating these categories are either being more effectively managed or are better suited to meet demand in the California market.

5) Product manager responsible for coffee is interested in sales of coffee across all the states. Which state has the highest coffee sales? Show it on a graph. What are the top 5 states for coffee sales?

The "Coffee Sales by State" bar graph gives a clear picture of how coffee sales perform in the US. California, with close to 24,000 units sold, stands out as being far better than any other state when it comes to coffee sales, as seen from the graph.

The five best states for coffee sales are:

- California about 24,500 units
- About 18,200 units in New York
- About 18,000 units in Illinois
- Around 16,500 units in Colorado
- Around 16,300 units in Texas



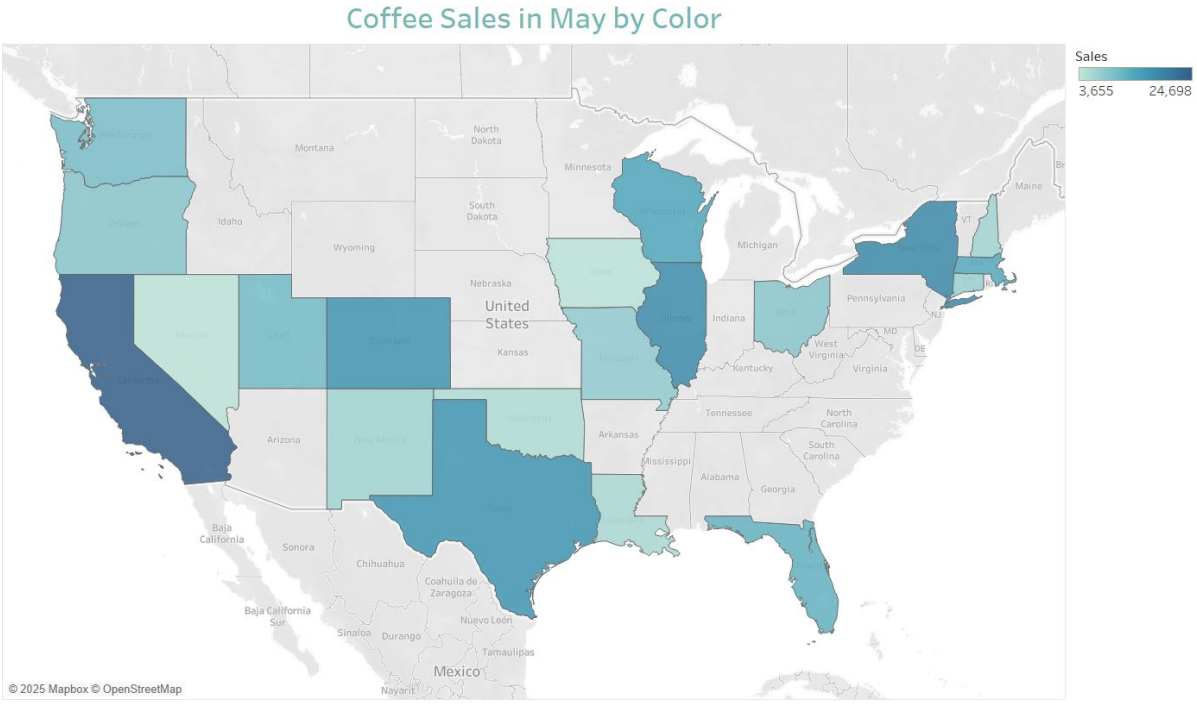
Sum of Sales for each State. The data is filtered on Product Type, which keeps Coffee.

These are followed by California by a huge lead and thus are the most profitable markets for coffee sales. This makes it highly likely to tap consumer demand in these segments through new products, new market development, or focused marketing.

6) Look at Profit by Regions (Market) by Product Type and Product. Can you identify a region with a negative Profit? What do you think is causing it? Which product and/or state is the main cause?

The deeper blue that shows over 24,000 sales for coffee sales is California as well as any region in the US, confirming California is at the top of the country, as usual. California is also a region of concern because earlier profit information (part of question 5) showed some of the coffee

products across the state—even Amaretto and Decaf Irish Cream—are losing money on high sales volume.



Map based on Longitude (generated) and Latitude (generated). Color shows sum of Sales. Details are shown for State. The data is filtered on Product Type, which keeps Coffee.

Despite increasing levels of sales, the mingling of low margins on certain lines of coffee products and high expenses is likely to be the cause for this poor profitability. Loss-leading products or special offers lowering profit rates are Decaf Irish Cream and Amaretto. The phenomenon is indicative of how crucial it is to regularly check profitability at the product level in major markets as well as keep a check on sales performance.

In conclusion, notwithstanding the strong sales performance, California is a huge state with some negative profit in the coffee product categories. Strategic price and product mix changes might be needed to enhance margin performance in this otherwise strong market.